

Business Model Canvas – MedTrack

Key Partners • Cloud provider (Microsoft Azure) • Hardware component suppliers (ESP32-C3, stepper, sensors) • Open Data providers (drug / interaction database) • Retirement / nursing homes (domovy dôchodcov) • Pharmacies & home-care / assisted-living organisations • App distribution platforms (Google Play, Web)	Key Activities • App + firmware development & maintenance • Accessibility-driven UX design & testing • Cloud backend operation & security (Firestore rules) • Hardware sourcing / assembly	Value Propositions • The senior never has to use the app: caregivers / family set up medications and monitor adherence, while the senior only does one simple physical step • Three equivalent low-effort ways to confirm a dose: in-app tap, NFC tag, physical DoseBuddy button • Reduced cognitive load: large high-contrast UI, status by colour + icon + text • Forgiving, configurable reminders & confirmation windows • Peace of mind for families via read-only caregiver access • Optional hardware dispenses one pill, multi-modal LED + buzzer feedback, works offline	Customer Relationships • Self-service with automated daily reminders • Caregiver-assisted setup & monitoring • Forgiving timing tailored to each routine	Customer Segments • Primary: elderly (65+) & motor-/cognitively-impaired patients on daily medication • Secondary: informal caregivers / family members • B2B: home-care agencies, assisted-living facilities, pharmacies
	Key Resources • Flutter codebase (Android + Web) • Firebase backend (Auth + Firestore) • DoseBuddy firmware & hardware design • Open Data drug reference dataset • Team development skills		Channels • Android app (full features) • Web app (reduced, demo via link) • DoseBuddy hardware companion • QR-code account sharing	
Cost Structure • Cloud hosting & database operations • Hardware bill of materials & assembly • Development & maintenance effort • Marketing & advertising (customer acquisition: family, care homes, pharmacies) • App store / distribution costs			Revenue Streams • Freemium app (core adherence free; premium caregiver / analytics features) • One-off DoseBuddy hardware sales (button & dispenser variants) • B2B licensing to care providers	
Social & Environmental Cost • Electronic waste from hardware devices & batteries • Energy / carbon footprint of cloud infrastructure • Handling of sensitive health data (privacy risk)			Social & Environmental Benefit • Better adherence → fewer avoidable hospitalisations & complications • Greater independence & dignity for elderly / disabled users • Reduced caregiver burden & reassura	